IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

EEE PART IV: MEng and ACGI

SUSTAINABLE ELECTRICAL SYSTEMS

Thursday, 7 May 2026, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : E. Spyrou, G. Strbac

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ANSWER ALL QUESTIONS

1. **Question on wind and solar PV plants**

a) A 15 MW wind turbine is installed at a site you manage. The wind turbine parameters are given in Table 1.1 and the density of air is 1.22 kg/m^3.

Table 1.1 Parameters of the wind turbine

Parameter	Value
Length of turbine blade	118 m
Cut-in speed	3 m/s
Cut-out speed	31 m/s
Rated speed	11 m/s
Gearbox efficiency	0.9
Generator efficiency	0.9

i) Calculate the total efficiency of the turbine. [2]

ii) Calculate the power coefficient of the turbine. [2]

iii) Provide the maximum total efficiency assuming an ideal aerodynamic design. [2]

iv) The forecast for tonight predicts wind speeds between 13 and 15 m/s. Calculate the range of wind power expected during these nighttime hours. [1]

v) The forecast for today's daylight hours predicts wind speeds between 9 and 11 m/s. Provide the range of wind power expected during daylight hours today. [1]

vi) Discuss how important wind speed forecast errors are during daylight and night hours today. [2]

b) Two solar PV plants are next to each other at latitude 50° in the UK. Plant A has a dual-axis tracking system. Plant B has a fixed tilt at 50° and no tracking.

i) Calculate the Sun's zenith angle at solar noon on the equinoxes. [1]

ii) Compare the electricity production from Plant A with the electricity production from Plant B at solar noon during the equinox. [2]

iii) Calculate the Sun's zenith angle at solar noon on the summer solstice. [1]

iv) Compare the electricity production from Plant A with the electricity production from Plant B at solar noon on the summer solstice. [2]

v) Rank the annual capacity factors of the two plants. Briefly explain the ranking. [2]

vi) A PV plant in the UK and an identical plant at the same latitude in Canada have capacity factors of 17% and 20%, respectively. Give two possible reasons why the plant in Canada has a higher capacity factor. [2]

2. **Question on security of supply**

a) State what the Capacity Outage Probability Tables (COPTs) specify and write the equation used to calculate Expected Energy Not Served (EENS), given energy not supplied E_i and capacity outage probability P_i for capacity state "i". [2]

b) Islanded Microgrid A contains a single generator with rated capacity of 3 MW and availability of 85%. Assuming a constant load of 2 MW, calculate the Expected Energy Not Served over 1-hour period. [1]

c) Islanded Microgrid B contains four identical generators, each of 2.25 MW capacity, and availability of 80%. Evening load profile is shown below in Fig. 2.1.

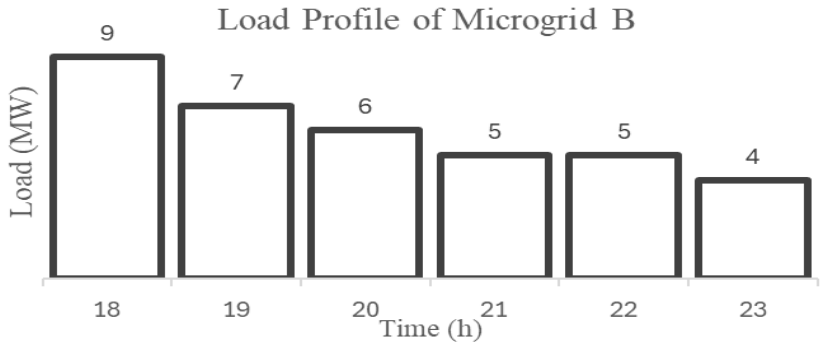


Figure 2.1 Load Profile of Microgrid B

i) Complete the table below (Table 2.1) and then calculate the total EENS over the six-hour period from 18:00 to 23:59. [3]

Table 2.1 Table to Fill for Microgrid B

Capacity in service (MW)	Probability	Energy not served in hours below (MWh)					
		18	19	20	21	22	23

ii) Using the results calculated in part (i), determine the 'effective generation capacity' of Microgrid B over the evening period, expressed in MW and as a percentage of the total installed generation capacity. Hence, determine the minimum number of generators required to achieve an effective capacity of at least 8 MW over this evening period. [4]

d) Microgrid C is connected to an external grid. This microgrid contains three distributed generators with parameters listed in table 2.2. For simplicity, assume that the annual load-duration curve of Microgrid C is represented by the straight line shown in Fig. 2.2.

Table 2.2 Generators in Microgrid C

Generator	Rated power (MW)	availability
G1	2	75%
G2	5	80%
G3	5	80%

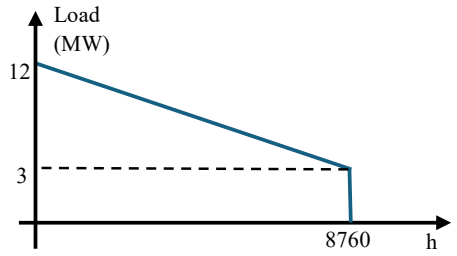


Figure 2.2 Load Duration Curve of Microgrid C

i) Assuming that no power is imported from the external grid, complete Table 2.3 and then calculate the Expected Energy Not Served. [6]

ii) Determine the minimum import capacity from the transmission line connecting Microgrid C to the external grid, such that the microgrid has an effective available capacity of at least 10 MW throughout the year. [4]

Table 2.3 Table to fill for Microgrid C

Capacity in service (MW)	Probability	Energy not served (MWh)

3. **Question on Low Carbon Heat**

a) Briefly describe the key factors that influence the demand for space heating in a domestic building. [2]

b) Figure 3.1 shows the total daily GB demand for domestic space and water heating plotted against average daily external temperature.

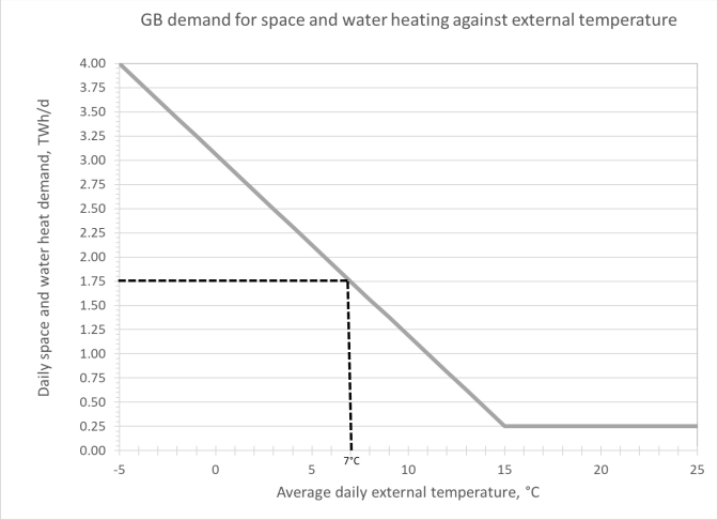


Figure 3.1 GB Demand for space and water heating against external temperature.

Assuming:

- 30 million domestic buildings
- Average building floor area of $100m^2$
- Space heating period is 200 days
- Water heating period is 365 days
- Average daily external temperature for the year of 7°C
- Average building internal temperature for the year of 17°C

Determine:

i) The average household annual water heat demand. [1]

ii) The average household annual space heat demand. [1]

iii) The average household Heat Loss Parameter. [2]

c) Climate change is estimated to result in an increase in GB average temperatures of 2°C for the heating period by 2050. Assuming a 30% average improvement in building insulation and no changes in other data:

i) Estimate the average household space heat demand in 2050. [1]

ii) Estimate the total GB household demand for space and water heating. [1]

iii) Plot the revised space and water heating demand for 2050 against average daily external temperature. [2]

d) Low carbon heating systems

i) Briefly describe an air source heat pump. [1]

ii) Briefly describe district heating and its main components. [1]

iii) Briefly describe hydrogen (with carbon capture and storage) heating. [1]

iv) Briefly describe Direct electric heating. [1]

v) Compare all heating systems described in i)-iv) above using the "spidergram" of Fig. 3.2 where "1" is ranked "Best" and "4" is the "Worst". The parameters chosen for comparison are:

- How easy is it to deploy the technology?
- How much does it cost to buy?
- How much does it cost to run?
- What are the CO2 emissions?
- What are the energy security implications?

As an example, gas is shown on the spider diagram. This technology is already deployed and so it scores high relative to others on both deployment and investment. However, it performs less well on running costs and badly on CO2 emissions. Finally, as the UK is largely dependent on gas imports it also scores badly on energy security.

Plot each of the 4 low carbon heating systems on the spider gram and explain your ranking rationale. [6]

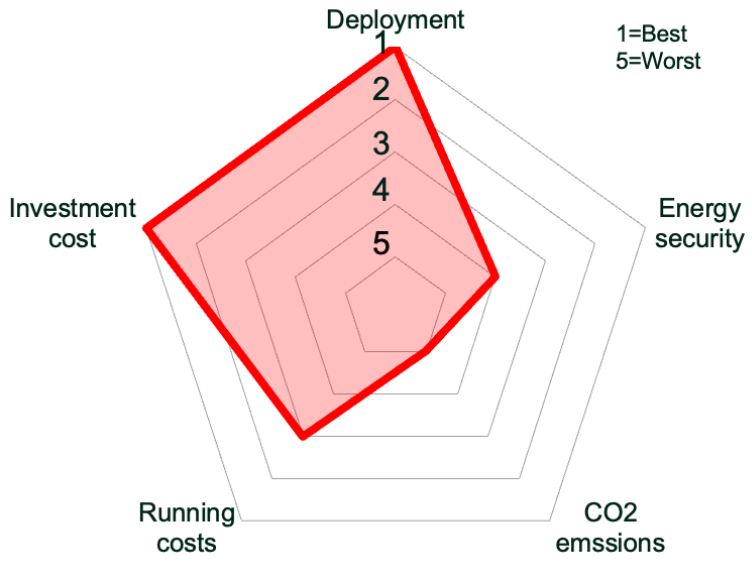


Figure 3.2 Spidergram

4. **Question on storage and electric vehicles**

a) A battery has power capacity 10 MW and usable energy capacity 40 MWh. It has 50% state of charge at 10:00 am and round-trip efficiency 80%.

i) Starting at 10:00 (am), the battery discharges for 12 minutes. Calculate the state of charge at 10:12. [2]

ii) The battery can start charging at 10:12. When will its state of charge be 50%? [2]

iii) What is the depth of this cycle? [1]

iv) Is this cycle a full cycle? [1]

v) How many cycles like the one described above the battery can complete in a year? [2]

vi) If the usable energy capacity changes to 20 MWh, what is the depth of the same cycle? [2]

b) Explain the concept behind the Vehicle-to-Grid (V2G) technology and discuss the challenges and opportunities associated with using V2G in low-carbon electricity systems. [3]

c) The baseline demand profile in a local distribution network during evening peak hours is shown in Figure 4.1. The Local Distribution Network Operator (DNO) needs to accommodate additional demand from 20 electric vehicles (EVs). The charging profile for a single EV is given in Figure 4.1.

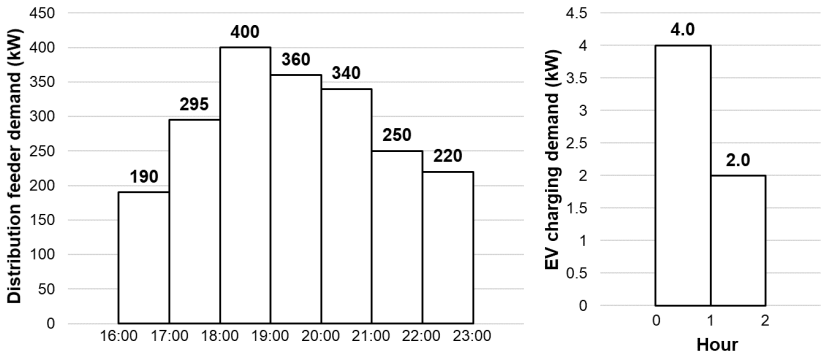


Figure 4.1 (left) Distribution feeder demand profile; (right) single EV charging demand profile

Answer the following questions:

i) What would be the new peak demand in the feeder if all EVs start charging at 6pm? Plot the new demand diagram with the EV demand included. [3]

ii) Assuming that each EV is flexible and can postpone their charging, what would be the earliest that all EVs can finish charging without increasing the total feeder peak demand? What would be the new EV charging schedule, i.e. how many EVs should start charging at which hour? Plot the new diagram for total feeder demand. (Assume that each EVs can follow its own charging schedule.) [4]